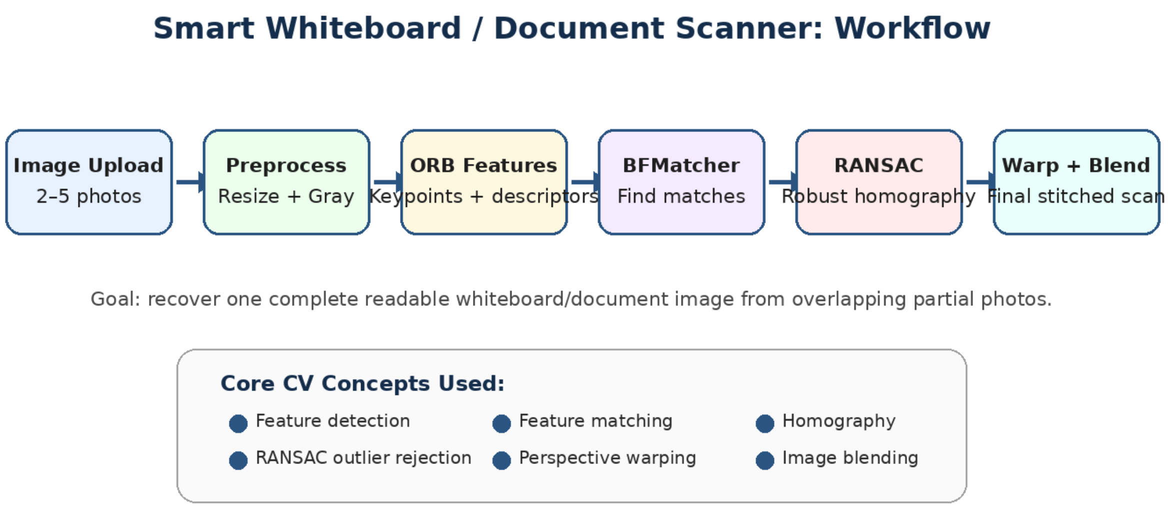
UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

Computer Vision Project Proposal

Smart Whiteboard / Document Scanner Using Image Stitching

A Practical Computer Vision Application for Digitizing Large Whiteboards and Documents



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1. Abstract

This project proposes the design and implementation of a Smart Whiteboard / Document Scanner using image stitching. The application will take two to five overlapping images of a whiteboard, classroom board, poster, map, or large document and merge them into one complete high-resolution image. The problem is common in classrooms and offices because large visual content often cannot be captured clearly in a single camera frame. As a result, users take multiple photographs, but those images remain separate and difficult to read as one complete document.

The proposed system will use classical Computer Vision techniques that are directly aligned with the course content. The main pipeline includes image acquisition, grayscale conversion, ORB keypoint detection, descriptor extraction, feature matching using BFMatcher, robust homography estimation using RANSAC, perspective warping, and final image blending. The project will be implemented in Python on Google Colab using OpenCV, NumPy, and Matplotlib. The expected outcome is a working prototype notebook that clearly shows all intermediate results, including input images, detected keypoints, matched features, warped images, and the final stitched image.

The project is suitable for undergraduate Computer Vision because it connects theory with a practical real-world application. It demonstrates feature-based alignment, projective transformation, and image stitching in a way that is useful for students, teachers, and professionals.

2. Introduction

Computer Vision is a field of Artificial Intelligence and image analysis that enables computers to interpret visual information from images and videos. A major task in Computer Vision is to recover useful information from visual data, such as edges, corners, objects, shapes, camera motion, and scene structure. Image stitching is one of the most practical examples of this field because it combines multiple overlapping images into a single wider or larger image.

In many educational settings, teachers explain lessons on large whiteboards. Students usually capture pictures of the board with a mobile phone. However, one picture may not cover the entire board. Even when it does, the text may become small, unclear, or distorted. The same issue appears when scanning large posters, engineering drawings, charts, maps, and office documents. A smart scanner that combines several close-up images can produce a clearer and more complete output.

This project focuses on a whiteboard/document scanning application. Instead of using deep learning, the system uses classical feature-based image stitching. This is useful because it helps understand the foundations of Computer Vision, especially feature detection, feature matching, homography estimation, and perspective transformations. These concepts are also used in panorama generation, augmented reality, camera calibration, and 3D reconstruction.

3. Problem Statement

Large whiteboards and documents are difficult to digitize using a single image. If the camera is too far away, the text becomes unreadable. If the camera is close, only a part of the board is captured. Users then take multiple photographs, but the images remain separate and do not form one continuous document. Manually combining images is time-consuming, inaccurate, and often produces misalignment.

The key challenge is to automatically identify which parts of the images overlap, align them geometrically, and merge them into a single output. The system must handle minor viewpoint changes, scale differences, rotation, and imperfect feature matches. Therefore, a robust image stitching approach is required. This project addresses the problem by using feature matching and RANSAC-based homography estimation to align overlapping images and generate a final stitched whiteboard/document scan.

4. Aim and Objectives

The main aim of this project is to develop a beginner-friendly Computer Vision application that automatically stitches overlapping images of a whiteboard or document into one readable image.

Objective	Explanation
Image acquisition	Allow the user to upload two to five overlapping images in Google Colab.
Preprocessing	Convert images into grayscale and resize them if required for efficient processing.
Feature detection	Use ORB to detect stable corners/keypoints in each image.
Feature matching	Use BFMatcher to find corresponding points between overlapping images.
Robust alignment	Use RANSAC to estimate homography while rejecting incorrect matches.
Image warping	Use perspective transformation to place images in a common coordinate system.
Final stitching	Combine warped images into a single output and display the stitched scan.
Evaluation	Check output quality using visual inspection, number of matches, and readability.

5. Literature Review / Technical Background

Image stitching is a well-known Computer Vision technique used to create panoramas and larger document images. The general idea is to find corresponding visual features in overlapping images and compute a transformation that maps one image onto another. Feature-based alignment is especially useful when images contain distinctive corners, text, diagrams, or repeated structures.

Feature detection methods identify important points in an image. ORB (Oriented FAST and Rotated BRIEF) is selected for this project because it is efficient, available in OpenCV, and suitable for real-time or beginner-level implementation. After detecting keypoints, the algorithm computes descriptors that summarize the local image pattern around each keypoint. These descriptors are matched between image pairs using BFMatcher.

A homography is a 3×3 projective transformation matrix that relates two views of the same plane. Since a whiteboard or paper document is approximately planar, homography is appropriate for aligning the images. However, feature matching can produce wrong correspondences. RANSAC is used to select the best transformation by repeatedly testing subsets of matches and rejecting outliers. This makes the stitching process more robust.

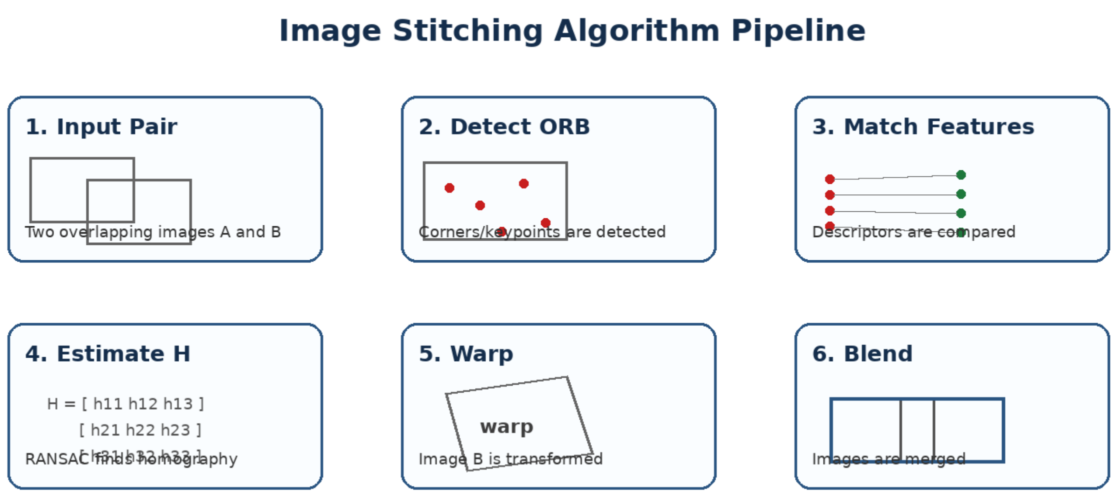

Figure 1: Image stitching algorithm pipeline used in the proposed project.

6. Proposed Methodology

The project will be implemented as a Google Colab notebook. The notebook will guide the user step by step, starting from image upload and ending with final stitched output. Every important step will display intermediate results so that the working of the algorithm is easy to understand.

Step	Process	Detailed Description
1	**Image Upload**	The user uploads two to five overlapping photographs of the same whiteboard/document.
2	**Read and Display**	Images are loaded using OpenCV and displayed using Matplotlib to verify the input.
3	**Grayscale Conversion**	Color images are converted to grayscale because ORB works efficiently on intensity patterns.
4	**ORB Feature Detection**	ORB detects corners and distinctive points in each image and computes descriptors.
5	**Feature Matching**	BFMatcher compares descriptors and finds likely corresponding points between images.
6	**Match Filtering**	The best matches are selected to reduce noisy correspondences.
7	**Homography Estimation**	RANSAC estimates a 3×3 homography matrix and removes outlier matches.
8	**Perspective Warping**	One image is warped into the coordinate system of another image.
9	**Stitching and Blending**	Images are placed on a shared canvas and combined into a single final output.
10	**Error Handling**	If there are too few matches or homography fails, the notebook shows a clear error message.

7. System Architecture and Diagrams

The system architecture is divided into three main layers: user layer, processing layer, and output layer. The user layer handles image upload and result viewing. The processing layer performs the main Computer Vision operations. The output layer displays intermediate visualizations and the final stitched image.

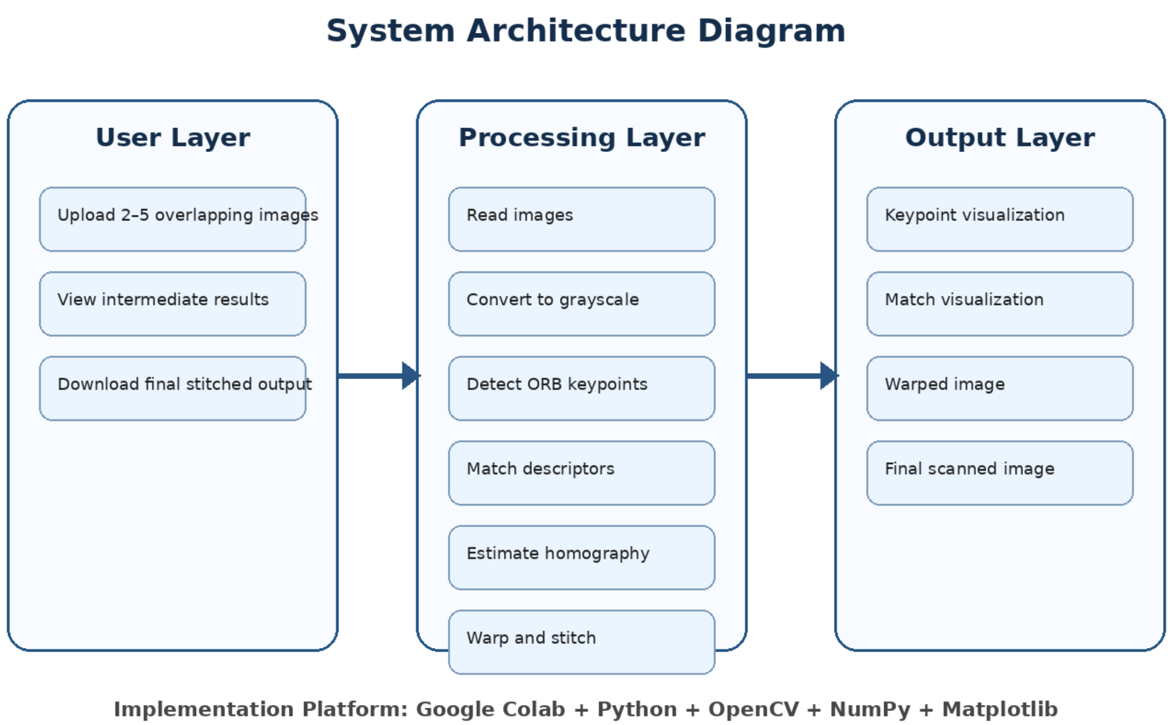

Figure 2: Proposed system architecture.

The workflow diagram below summarizes the complete movement of data from input photographs to final stitched scan. It also highlights the main Computer Vision concepts used in this project.

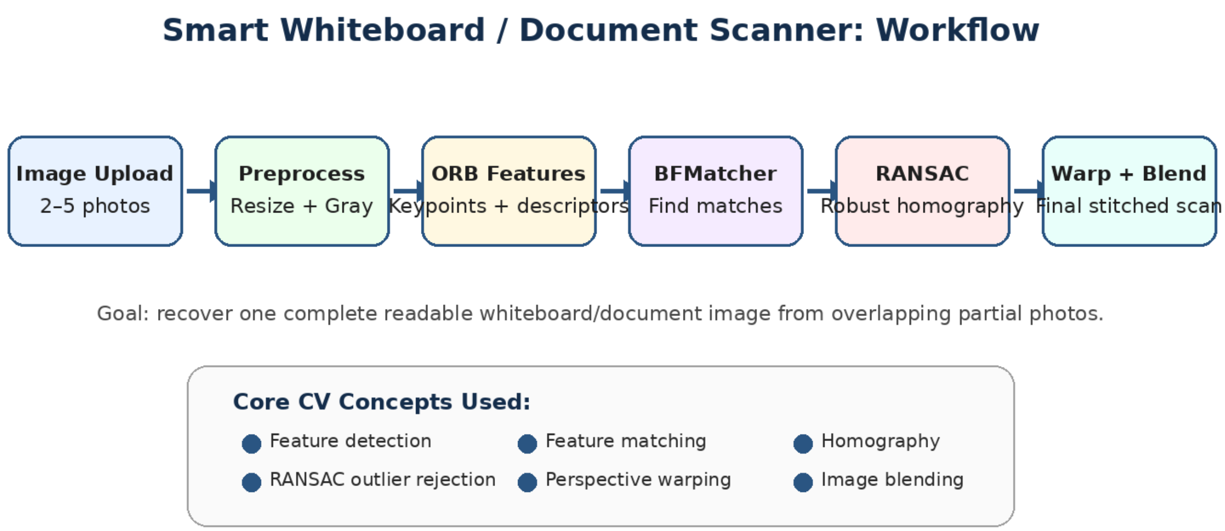

Figure 3: Complete project workflow.

7.1 Example Images Required for the Project

The application requires images of the same scene with overlap. Good input images should contain clear text, corners, diagrams, or other distinctive patterns. Consecutive images should overlap by around 30–50%. Images should not be blurry, unrelated, or captured from extremely different viewpoints.

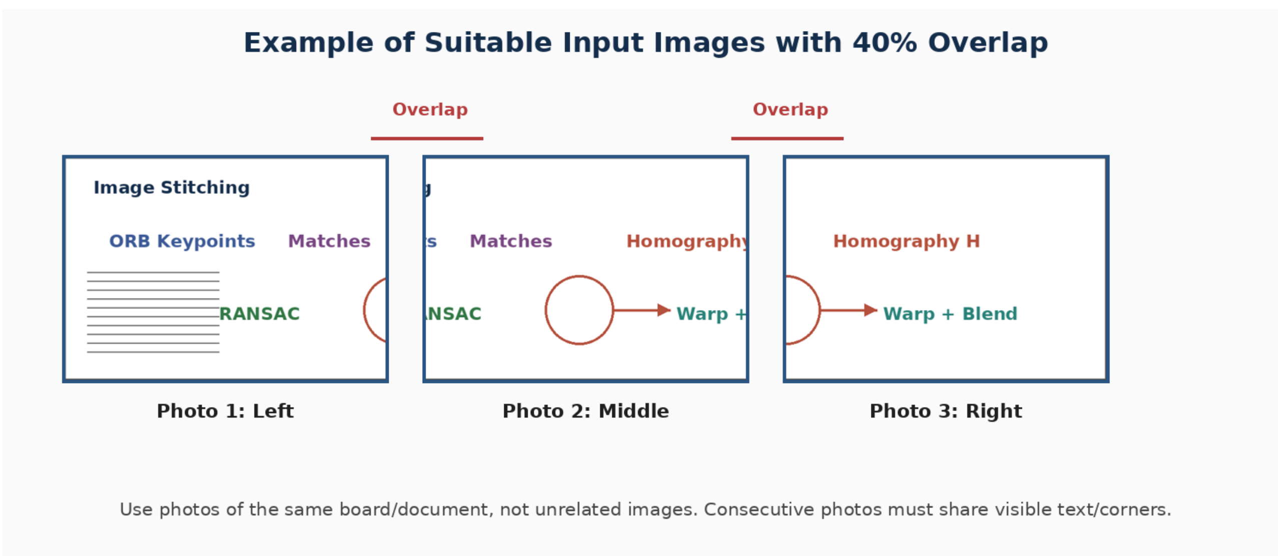

Figure 4: Example of suitable input photos for whiteboard/document stitching.

8. Tools, Technologies, Hardware and Software Requirements

Category	Requirement	Purpose
Programming Language	Python	Main implementation language for the project.
Computer Vision Library	OpenCV	Image processing, ORB, matching, homography, warping, and stitching.
Numerical Library	NumPy	Matrix operations and image array handling.
Visualization	Matplotlib	Display input images, keypoints, matches, and output.
Platform	Google Colab	Cloud-based notebook for running the project without local setup.
Hardware	Laptop/PC + Mobile Camera	Capture images and run the notebook.
Memory	Minimum 4GB, Recommended 8GB	Enough to process multiple images.

9. Feasibility Analysis

The project is feasible for an undergraduate Computer Vision course because it uses well-established algorithms and open-source tools. It does not require expensive hardware or a large training dataset. The main requirement is a small set of overlapping images captured by a mobile phone camera.

Feasibility Type	Rating	Explanation
Technical Feasibility	**High**	OpenCV provides ORB, BFMatcher, homography estimation, and warping functions.
Economic Feasibility	**Very High**	The project uses free tools such as Google Colab, Python, NumPy, and OpenCV.
Operational Feasibility	**High**	The system is easy to use because users only upload images and run cells.
Schedule Feasibility	**High**	The project can be completed within 6–8 weeks using a modular approach.
Academic Feasibility	**Very High**	It directly supports Computer Vision topics such as features, matching, and geometry.

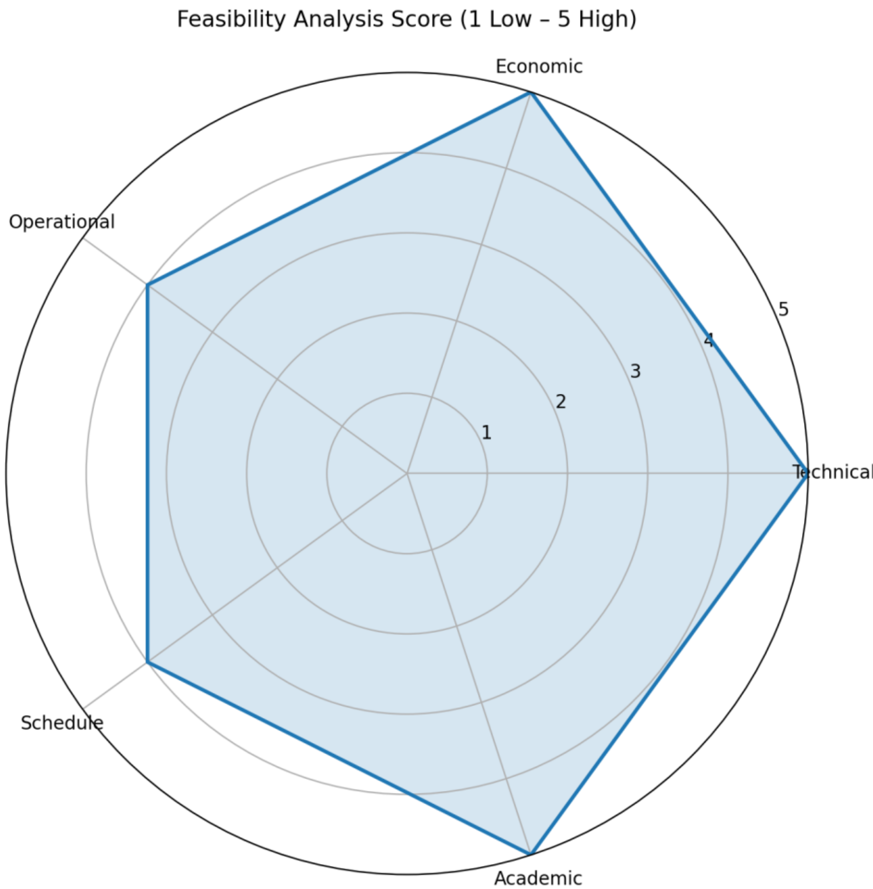

Figure 5: Feasibility analysis visualization.

10. Expected Results and Evaluation Criteria

The expected result is a final stitched image that combines all uploaded overlapping photos into one complete scan. The output should be readable, aligned, and visually continuous. The system should also show intermediate images so that the algorithm can be evaluated step by step.

Evaluation Criterion	Expected Standard
Overlap detection	The system should find enough matches between consecutive images.
Homography quality	RANSAC should produce a valid transformation matrix.
Visual alignment	Text and diagrams should align naturally after warping.
Readability	The final stitched document should be easy to read.
Error handling	The notebook should warn the user when images have insufficient overlap.
Intermediate outputs	Keypoints, matches, warped image, and final result should be displayed.

11. Applications and Beneficiaries

Beneficiary	Benefit
Students	Can save complete classroom whiteboard notes in one readable image.
Teachers	Can preserve lecture boards and share them with students.
Office Users	Can digitize meeting boards, project plans, posters, and charts.
Researchers	Can scan large diagrams, maps, or laboratory boards.
General Users	Can create panoramas or combine large documents without special scanning hardware.

12. Project Timeline / Gantt Chart

The following Gantt chart shows an 8-week project plan. Each phase is designed to build on the previous phase, from literature review to final presentation.

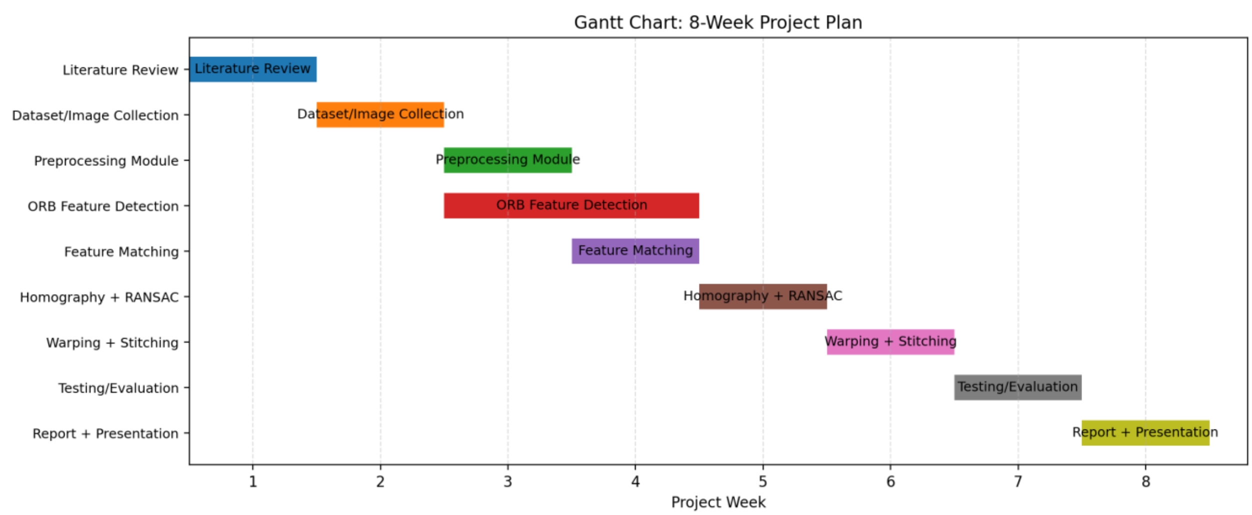

Figure 6: Proper Gantt chart for project planning.

Week	Task	Deliverable	Status/Remarks
1	Literature Review	Understanding image stitching concepts	Foundation phase
2	Dataset/Image Collection	Overlapping sample images	Use whiteboard/document photos
3	Preprocessing + ORB	Keypoint detection output	Intermediate visualization
4	Feature Matching	Match visualization	BFMatcher results
5	Homography + RANSAC	Robust transformation matrix	Outlier rejection
6	Warping + Stitching	Initial stitched output	Core implementation
7	Testing + Evaluation	Improved output and error handling	Quality testing
8	Documentation + Presentation	Final report and demo	Submission phase

13. Risk Analysis and Mitigation Plan

Risk	Impact	Mitigation
Insufficient overlap between images	Feature matching may fail	Capture images with 30–50% overlap.
Blurry input images	Keypoints may not be detected properly	Use clear images and stable camera position.
Poor lighting/reflection	Text and features may be hard to detect	Use uniform lighting and avoid glare.
Wrong feature matches	Images may align incorrectly	Apply RANSAC and filter matches.
Large image size	Processing may become slow	Resize images before processing.
Non-planar scene	Homography may not work perfectly	Focus on planar boards/documents.

14. Limitations and Future Enhancements

The proposed system works best when the scene is mostly planar and images have clear overlap. It may fail if the images are unrelated, highly blurred, too dark, or captured from extreme viewpoints. Since the project uses classical Computer Vision, it may be less robust than modern deep learning approaches in highly challenging scenes.

Future enhancements can include automatic cropping, seam optimization, exposure correction, mobile application deployment, deep-learning-based feature matching, automatic document enhancement, perspective correction for individual images, and OCR for extracting text from the final stitched output.

15. Conclusion

The Smart Whiteboard / Document Scanner using Image Stitching is a practical and meaningful Computer Vision project. It solves a real problem faced by students and professionals while demonstrating important course concepts such as ORB feature detection, descriptor matching, RANSAC, homography estimation, image warping, and image stitching. The project can be implemented in Google Colab using Python and OpenCV and can be demonstrated with simple overlapping photographs of a whiteboard or document. The final output will be a stitched high-resolution image that preserves large visual content in a single readable form.

16. References

R. Szeliski, Computer Vision: Algorithms and Applications, Springer, Chapter 9: Image Stitching.

R. Hartley and A. Zisserman, Multiple View Geometry in Computer Vision, Cambridge University Press, Chapters 4 and 13: Homographies and Projective Transformations.

D. Forsyth and J. Ponce, Computer Vision: A Modern Approach, Chapters on image features, fitting, and multiple-view geometry.

R. C. Gonzalez and R. E. Woods, Digital Image Processing, Pearson, Chapters on image fundamentals, filtering, and segmentation.

OpenCV Documentation, ORB, BFMatcher, findHomography, warpPerspective, and Stitcher modules.

Footer / Source Validation: The report content is based on the provided Computer Vision knowledge base books: Szeliski (Image Stitching and Feature-Based Alignment), Hartley & Zisserman (Homography and Multiple View Geometry), Forsyth & Ponce (Image Features and Fitting), and Gonzalez & Woods (Digital Image Processing fundamentals).